HAFIZ NAUMAN BAIG

PhD Research Scholar (Zoology / Fisheries & Aquaculture)

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Academic Portfolio: **naumanbaig.me** | Applied Research Platform: **fishwaterlog.com**

1. RESEARCH INTERESTS

Fish environmental and thermal stress physiology, sustainable intensive aquaculture, molecular growth regulation, oxidative stress biomarkers, fish histopathology, transcriptomic profiling, brain plasticity, and innovative dietary prophylactic strategies (e.g., natural antioxidants) to optimize fish health, welfare, and production in modern recirculating aquaculture systems (RAS).

2. EDUCATIONAL QUALIFICATIONS

Degree / Certificate	Institution / University	Duration / Year	Performance / Status
Ph.D. Zoology (Fisheries & Aquaculture)	Department of Zoology, Faculty of Biological Sciences, Quaid-i-Azam University (QAU), Islamabad, Pakistan	Spring 2024–Present	**In Progress (Coursework: 18 Cr Completed)**
M.Phil. Zoology (Fisheries & Aquaculture)	Institute of Molecular Biology & Biotechnology (IMBB), The University of Lahore (UOL), Lahore, Pakistan	2021–2023	CGPA: 3.73 / 4.00 (First Division, 80.33%)
B.S. (4-Year) Zoology (Specialization: Fisheries)	Govt. College of Science, Wahdat Road, Lahore / University of the Punjab (PU), Lahore, Pakistan	2013–2017	First Division
Hifz-ul-Quran	Madrasa-tul-Madina, Lahore, Pakistan	2007	Sanad-e-Takmeel

3. M.PHIL. RESEARCH PROJECT

Title: "Physiological and Protective Role of Dietary Vitamin C Against ZnSO4-Induced Toxicity in Nile Tilapia (*Oreochromis niloticus*)"

Supervisors: Miss Maham Mazher & Prof. Dr. Sikandar Hayat (IMBB, UOL)

Summary & Core Methodology: Investigated the oxidative stress, heavy-metal induced bioaccumulation, biochemical alterations, and corresponding tissue histopathology caused by sub-lethal zinc sulfate (ZnSO4) exposure in juvenile Nile Tilapia. The study successfully demonstrated that therapeutic dietary ascorbic acid (Vitamin C) supplementation significantly ameliorates renal damage, muscle toxicity, and oxidative biomarkers. Key methodologies utilized included hematological profiling, colorimetric biochemical assays, antioxidant enzyme evaluation (SOD, CAT, TAC), and advanced liver/kidney microtomy and histopathology.

4. PROFESSIONAL RESEARCH & WORK EXPERIENCE

Fisheries Research Assistant | Department of Fisheries, Government of Punjab (Chakwal, Pakistan)

Duration: 09-07-2019 to 31-12-2020 (Approx. 1.5 Years) | Project: "Studies on Economic Analysis of Cage Fish Culture in Potohar Region"

- Conducted extensive cage culture operations, feeding regime optimization, and biological sampling in major freshwater reservoirs.
- Monitored water quality parameters (Dissolved Oxygen, pH, Temperature, Ammonia, Hardness) to maintain optimal environmental standards.
- Assisted in aquatic disease surveillance, fish parasite identification, and general health management protocols of intensive systems.
- Collected, curated, and analyzed fish biomass and growth metrics to draft monthly operational productivity reports for administrative reviews.

5. PEER-REVIEWED PUBLICATIONS & REGISTRATIONS

1. **Baig, H. N.**, Mazher, M., & Hayat, S. "Biochemical and histopathological studies on the ameliorative effect of vitamin C on renal and muscle tissues of Nile tilapia (*Oreochromis niloticus*) exposed to zinc sulfate." (*Manuscript Under Review / Submitted*).

2. **Baig, H. N.** (2026). Effects of Environmental Enrichment on Neuroendocrine Stress, Brain Plasticity, and Post-Mortem Muscle Quality in Teleost Fish under Thermal and Harvest Stress: A Systematic Review and Meta-Analysis Protocol. *Open Science Framework (OSF Registries)*. Persistent Identifier: **https://osf.io/kem4t/** (PRISMA-P & CAMARADES Compliant; Registered under Private Embargo until Sep 2029).

6. TECHNICAL & LABORATORY SKILLS

Technical Domain	Specific Methodologies & Core Competencies
Aquaculture & Physiology	Fish husbandry and Recirculating Aquaculture Systems (RAS) operations, water biochemistry monitoring, heavy-metal toxicity profiling, biological bio-assays, tissue micro-dissection, and animal health sampling.
Molecular Biology	Organelle/Tissue RNA extraction (TRIzol), cDNA synthesis, targeted RT-qPCR gene expression profiling, colorimetric enzymatic assays, ELISA diagnostics, and biochemical histopathology (microtomy).
Analytical & Soft Skills	Advanced biostatistical modeling in R-Studio (Multivariate Meta-Analysis [**metafor**], Linear Mixed Models [LMM], Bayesian ZOIB models [**brms**]), screening via **Rayyan**, SPSS, MS Excel, scientific synopsis/manuscript writing, and academic presentations.
Applied Web & Telemetry	Lead Developer & Editor — FishWaterLog (**fishwaterlog.com**): Designed diagnostic aquatic chemistry calculators, biofiltration tracking models, and science communication frameworks for recirculating aquaculture telemetry.

7. PROFESSIONAL CERTIFICATIONS & TRAININGS

• **Event Organizer & Presenter:** "Innovating for Tomorrow: Fish Health Management and Genetic Tools for Sustainable Fisheries" — International Symposium-cum-Workshop organized by Quaid-i-Azam University (QAU), Pakistan Fisheries Society, and HEC (January 2025).

• **FAO eLearning Academy (United Nations):** "Emergency Preparedness for Aquatic Disease Outbreaks" (December 2025).

• **FAO eLearning Academy (United Nations):** "Aquaculture Breeding and Genetics" (December 2025).

• **Tawakkal Fish Hatchery & Farms:** "100 Days Shrimp Farming Training Program" (August – November 2025).

• **Certificate of Completion:** "Ocean Governance: Sustainable Use of Ocean Resources" — International Ocean Institute (IOI) Ocean Academy (**September–October 2025**).

• **Government Professional Training:** "Extension, Conservation & Computer Skills" — Directorate of Fisheries R&T, Government of Punjab, Pakistan (February 2020).

8. ACADEMIC & PROFESSIONAL REFERENCES

Prof. Dr. Kiran Afshan
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